



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°6_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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NOM: PÉTIOTE

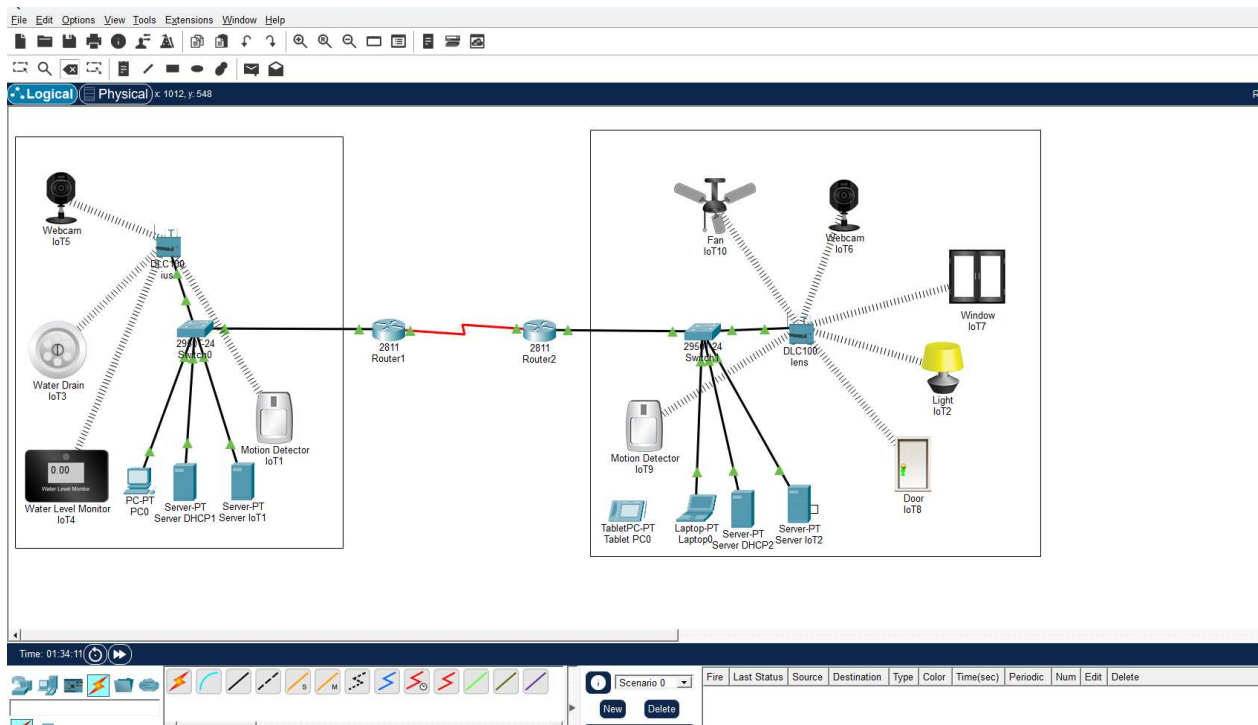
NIVEAU: L3

DATE: 04/05/2026

L'objectif de ce TD est de:

1. Le contrôle à distance
2. La communication entre objets
3. La sécurité du réseau IoT

Conception et configuration d'un réseau Smart Home (IoT)



Configuration des routeurs

R1

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

```

Ctrl+F6 to exit CLI focus

R2

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)## Configuration de l'interface srie
      ^
% Invalid input detected at '^' marker.

Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.168.1.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Configuration de l'interface FastEthernet0/0 pour Smarthome
      ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

```

Ctrl+F6 to exit CLI focus

Routage entre les deux routeurs (OSPF)

R1

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#do wr
00:08:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on
Serial0/2/0 from LOADING to FULL, Loading Done

Building configuration...
[OK]
Router(config-router)#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

 Top

R2

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

 Top

Configuration des servers DHCP pour attribuer des IP automatiquement

Server DHCP1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameserverPool

Default Gateway192.168.2.1

DNS Server0.0.0.0

Start IP Address19216823

Subnet Mask:2552552550

Maximum Number of Users :25

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.2.1	0.0.0.0	192.168.2.3	255.255.255.0	25	0.0.0.0	0.0.0.0

Top

Exemple

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE90:C5B2

IPv6 Gateway

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

DHCP 2

Server DHCP2

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameserverPool

Default Gateway192.168.3.1

DNS Server0.0.0.0

Start IP Address19216833

Subnet Mask:2552552550

Maximum Number of Users :25

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.3.1	0.0.0.0	192.168.3.3	255.255.255.0	25	0.0.0.0	0.0.0.0

Top

Configuration des servers lo

Server IoT1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

Registration Server

This service runs on top of the HTTP or HTTPS service.

Service

On

Off

Delete

Top

Server IoT2

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

Registration Server

This service runs on top of the HTTP or HTTPS service.

Service

On

Off

Delete

☐ Top

ius

Physical

Config

GUI

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Global Settings

Display Name

ius

Device Clock:

00:19:44 Mon Mar 1 1993 UTC

☐ Top

ius

Physical **Config** GUI Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Wireless Settings

SSID: admin

2.4 GHz Channel: 6 - 2.437GHz

Coverage Range (meters): 250.00

Authentication:

- ☒ Disabled
- ☐ WPA-PSK
- ☐ WPA
- ☐ WEP
- ☐ WPA2-PSK
- ☐ WPA2

WEP Key:

PSK Pass Phrase:

RADIUS Server Settings:

IP Address:

Shared Secret:

Encryption Type: Disabled

☐ Top

Exemple d'un appareil

IoT5

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

GigabitEthernet3

Bluetooth

Global Settings

Display NameIoT5

Serial NumberPTT0810NBV3

InterfacesWireless0

Gateway/DNS IPv4

☒ DHCP

☐ Static

Default Gateway192.168.25.1

DNS Server0.0.0.0

Gateway/DNS IPv6

☒ Automatic

☐ Static

Default Gateway

DNS Server

IoT Server

☒ None

☐ Home Gateway

☐ Remote Server

Server Address

User Name

Password

Connect

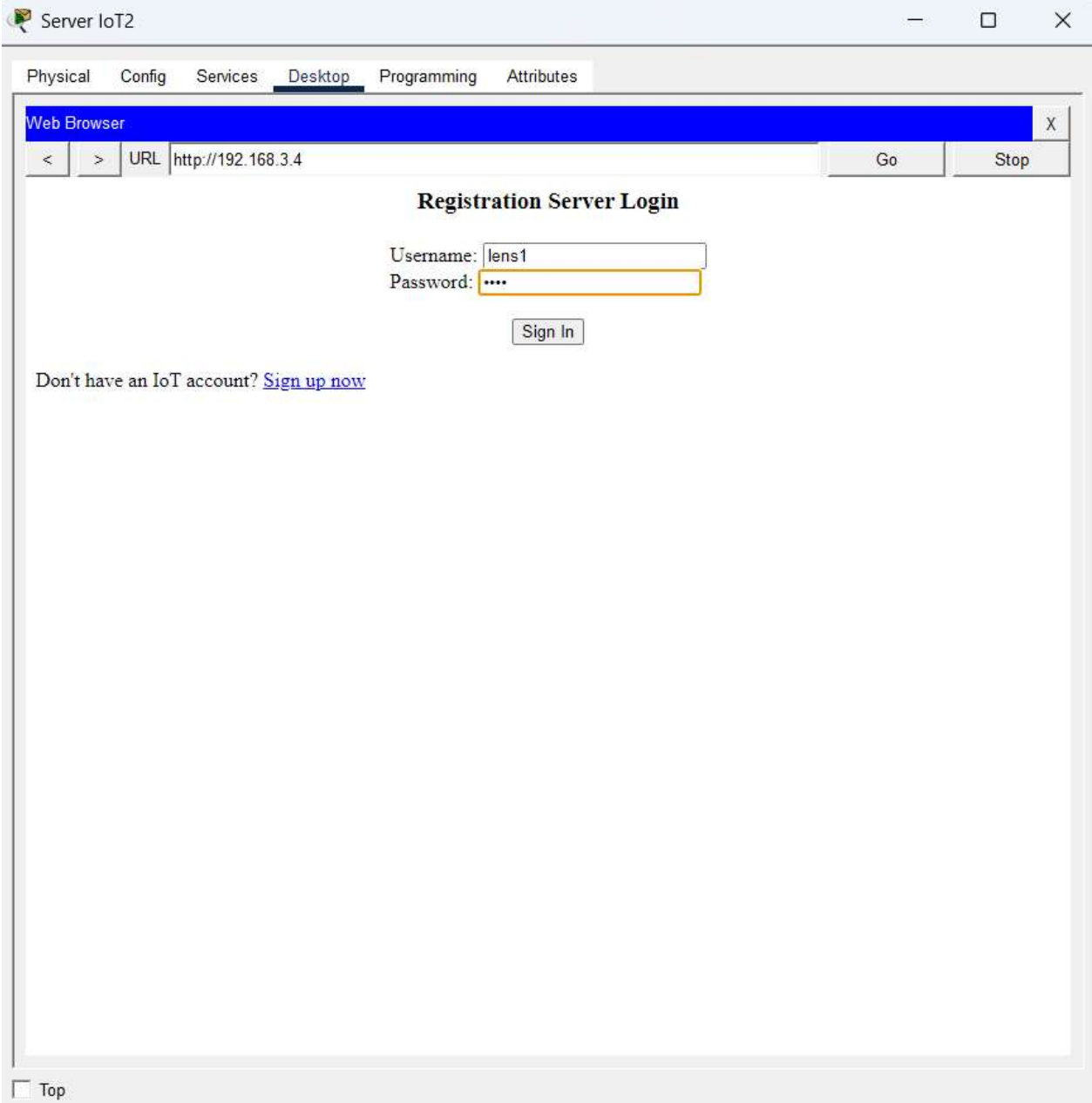
Device Clock: 00:24:42 Mon Mar 1 1992 UTC

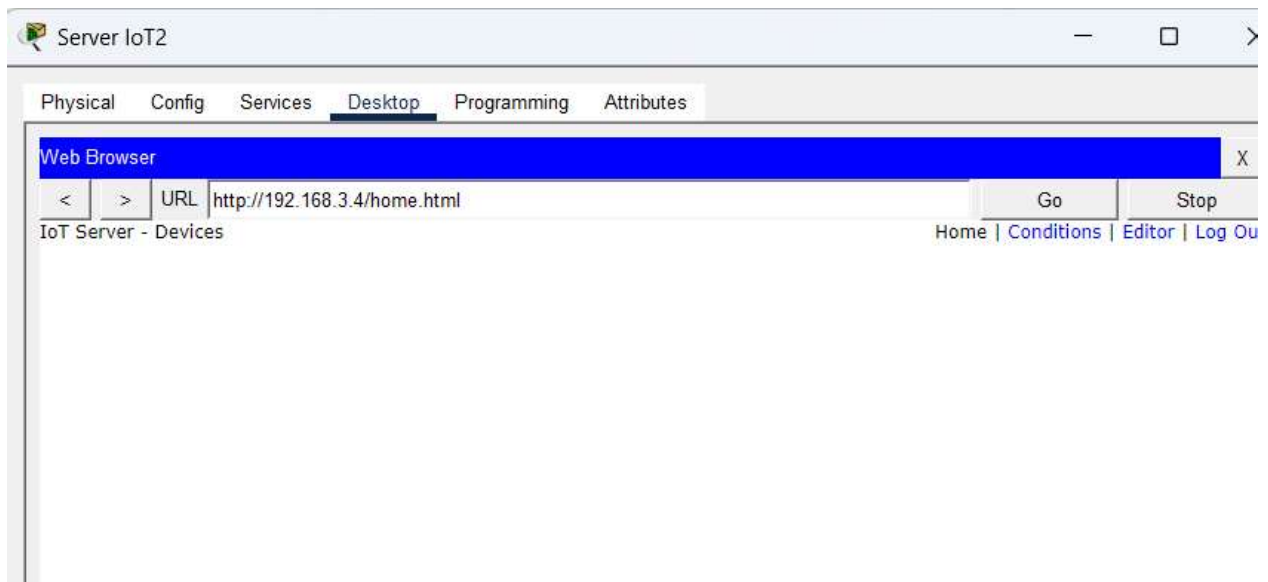
☐ Top

Advanced

Registration au server

IoT 1





Ajout des IoT sur le serveur

Exemple de configuration d'un appareil

IoT1

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Global Settings

Display NameIoT1

Serial NumberPTT0810SHMS

InterfacesWireless0

Gateway/DNS IPv4

☒ DHCP

☐ Static

Default Gateway192.168.25.1

DNS Server0.0.0.0

Gateway/DNS IPv6

☒ Automatic

☐ Static

Default Gateway

DNS Server

IoT Server

☐ None

☐ Home Gateway

☒ Remote Server

Server Address192.168.2.4

User Namelens

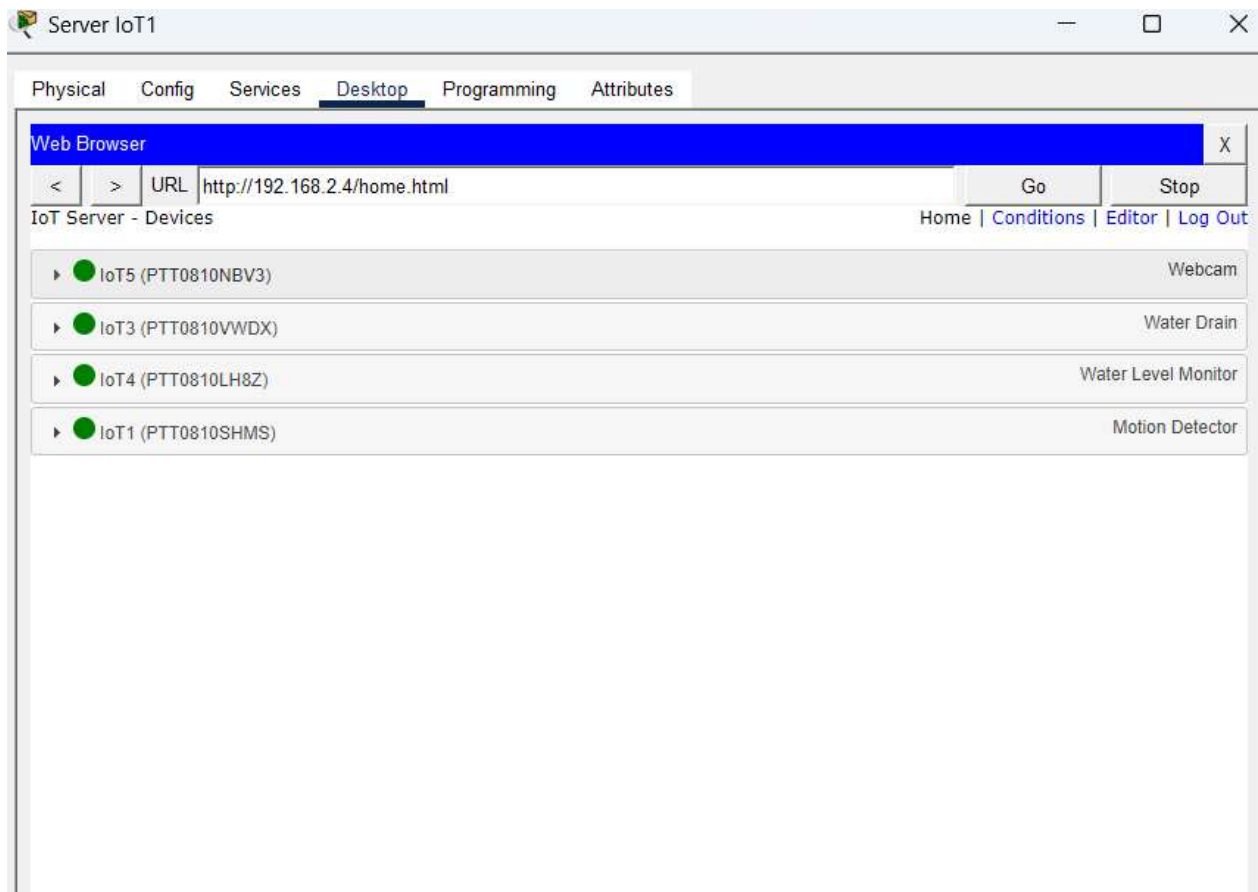
Password1234

Refresh

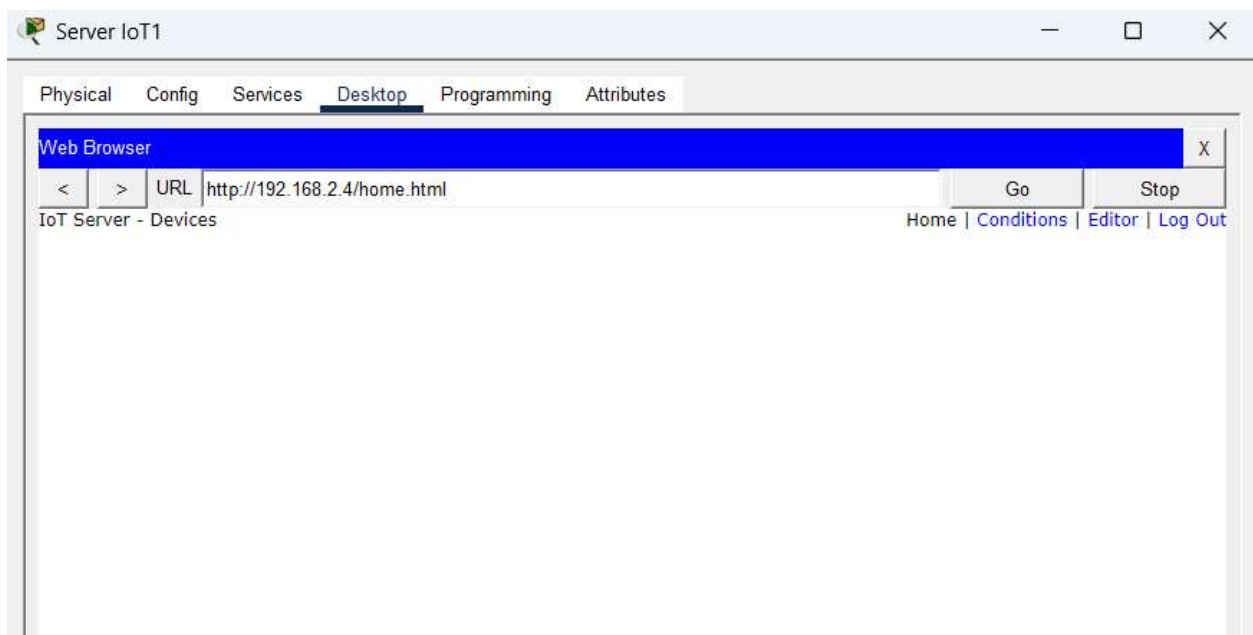
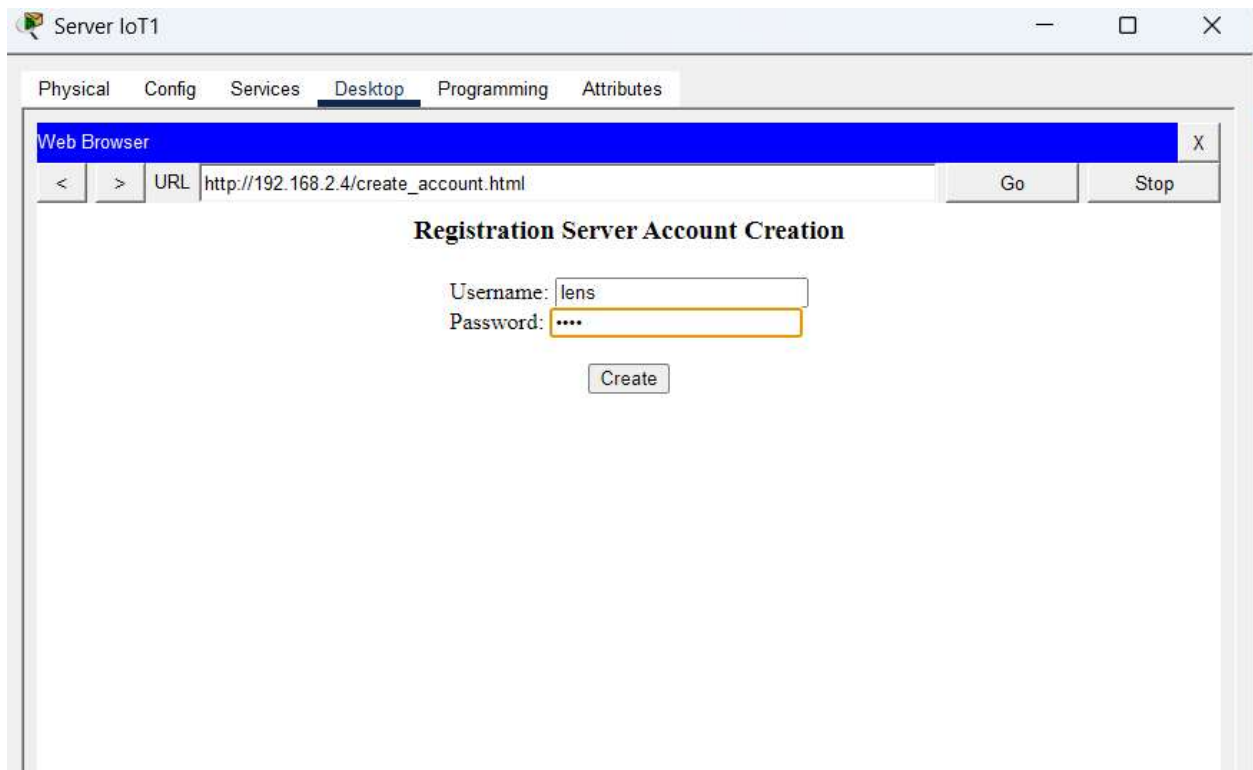
Device Clock: 00:45:44 Mon Mar 1 1993 UTC

☐ Top

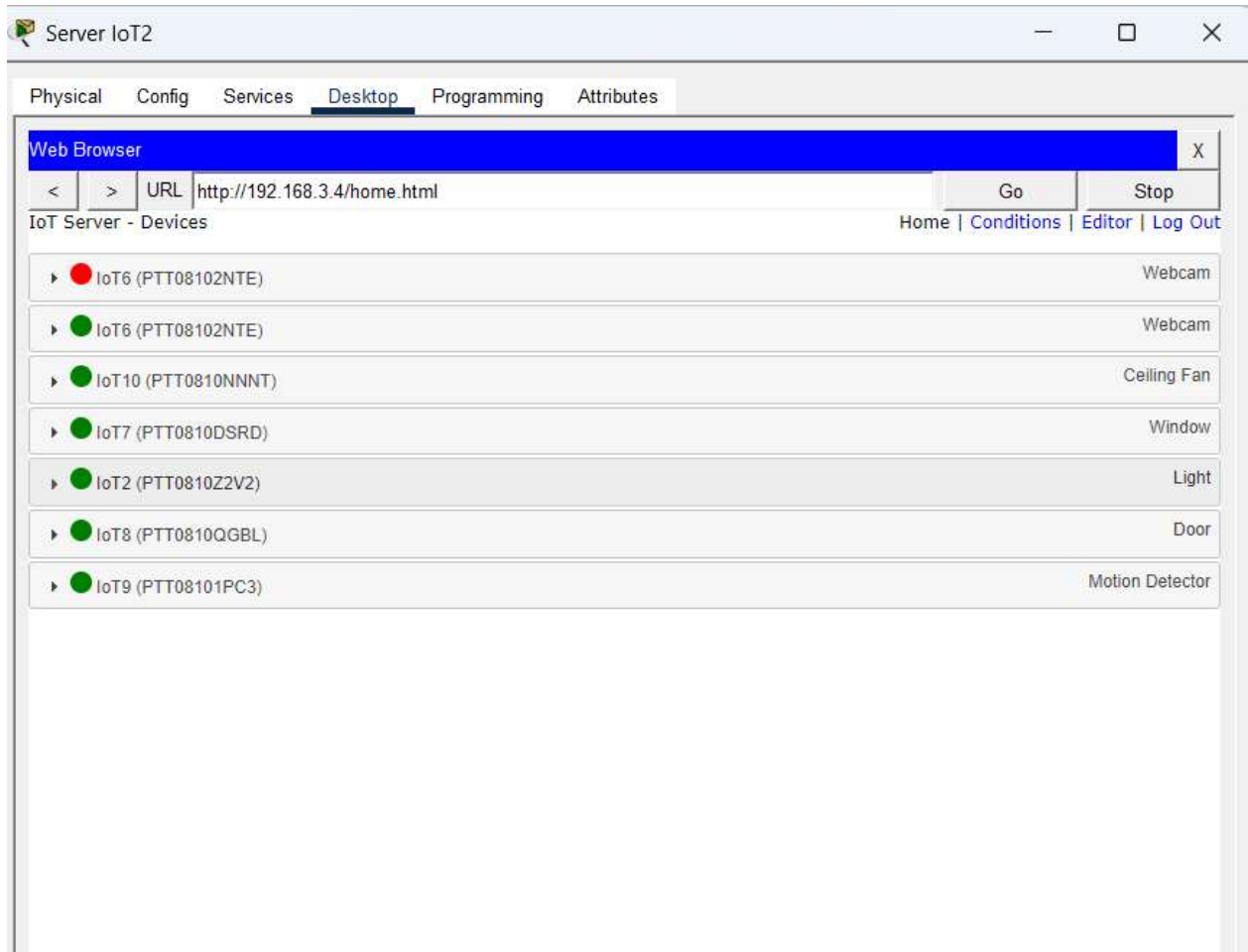
Advanced



IoT 2

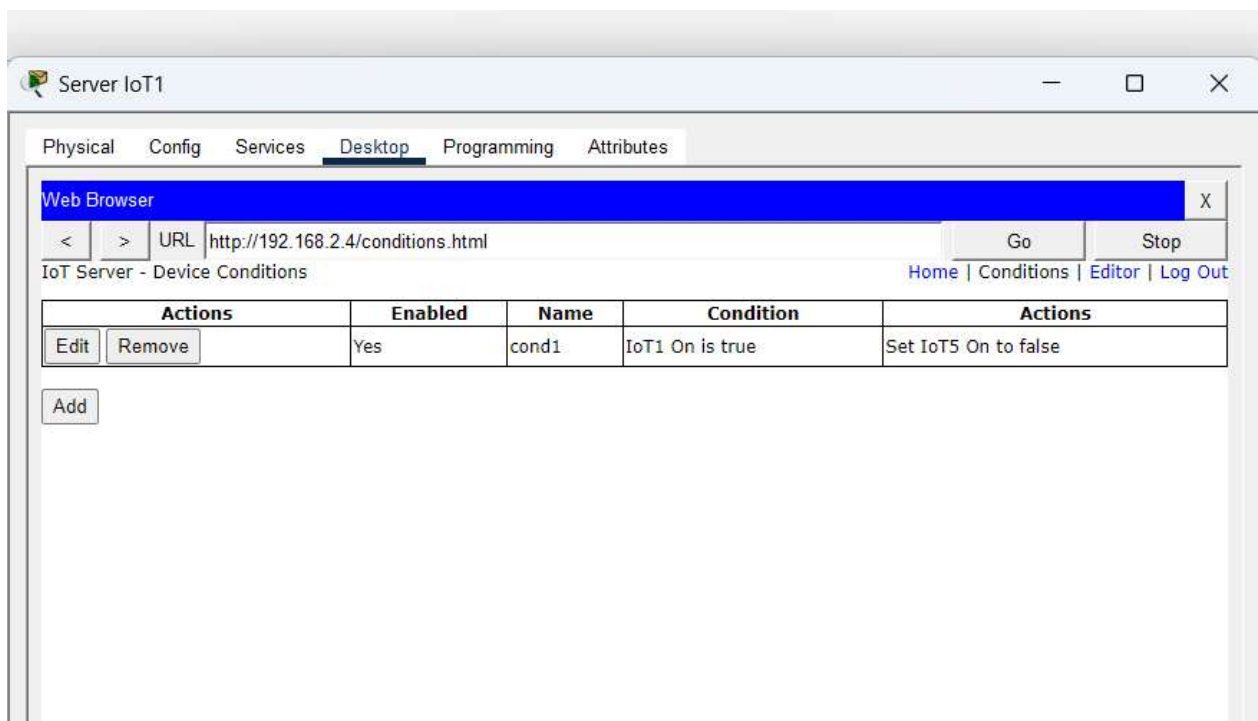
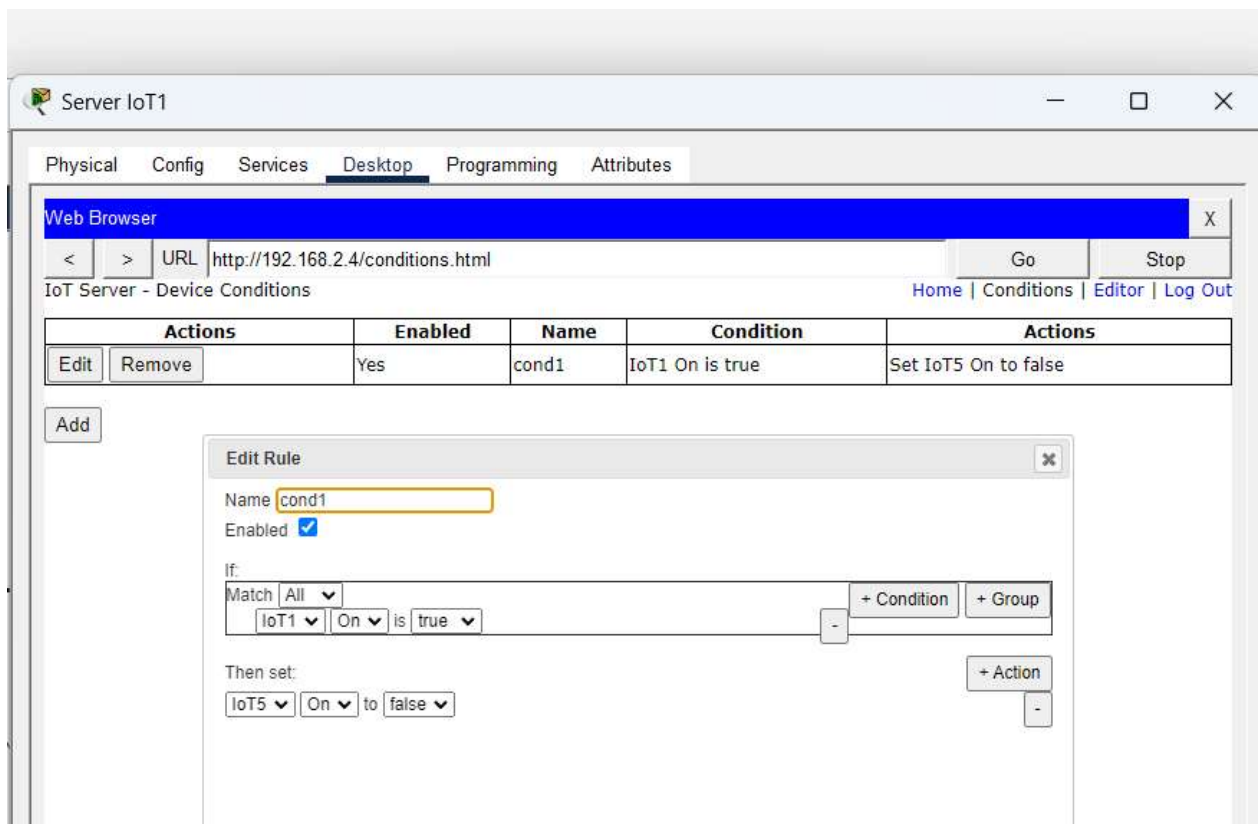


Ajout des IoT sur le serveur



Automatisation (IoT Rules)

IoT 1



Notre condition: Si le motion detector est ouvert, le camera s'eteint.

IoT 2

The screenshot shows the 'Server IoT2' application window with the 'Desktop' tab selected. A 'Web Browser' window is open, displaying the URL 'http://192.168.3.4/conditions.html'. Below the browser, the 'IoT Server - Device Conditions' page is visible, featuring a table of conditions and an 'Add' button.

Actions		Enabled	Name	Condition	Actions
Edit	Remove	Yes	cond1	IoT7 On is true	Set IoT10 Status to Off

An 'Add' button is located below the table. An 'Edit Rule' dialog box is open, showing the configuration for 'cond1'. The dialog includes fields for 'Name' (cond1), 'Enabled' (checked), and a rule definition. The rule is defined as: 'If: Match All IoT7 On is true' followed by a minus sign and a plus sign. The 'Then set:' section shows 'IoT10 Status to Off' followed by a plus sign and a minus sign. The dialog has 'OK' and 'Cancel' buttons at the bottom.

Si la fenetre est ouvert, le ventilateur doit s'eteindre.

Condition 2

Add Rule

Name

Enabled ☒

If:

Match

All

IoT8

Open

 is

true

-

+ Condition

+ Group

Then set:

IoT2

Status

 to

Off

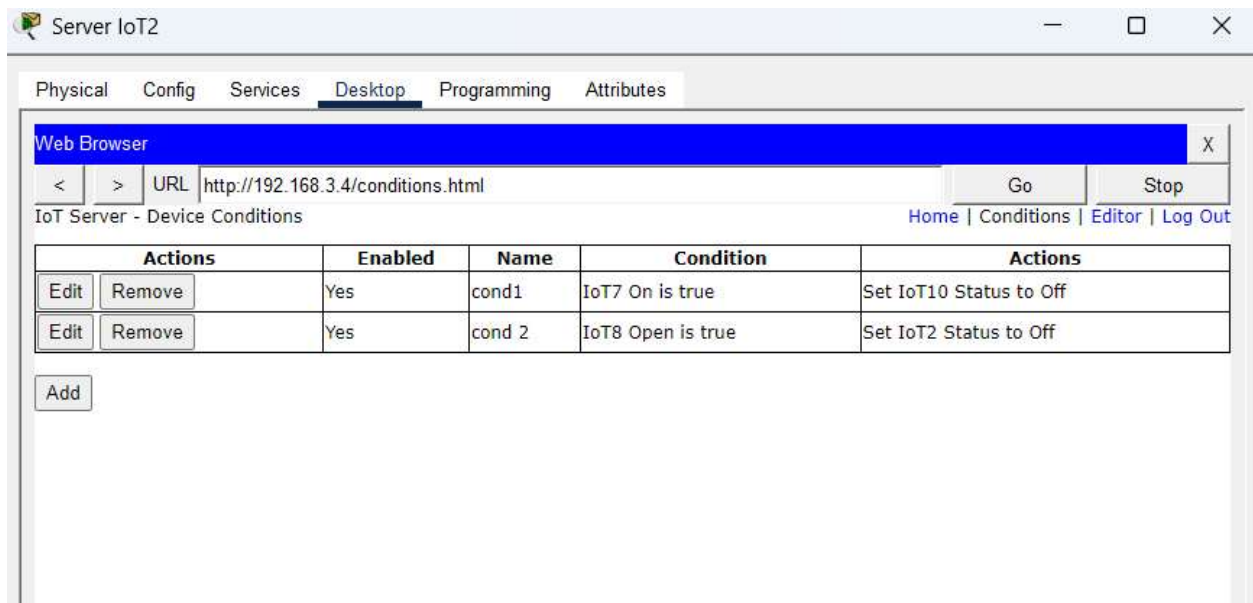
-

+ Action

OK

Cancel

Si la porte est ouvert la lampe doit s'eteindre



Coclusion.

Ce TD m'a permis de savoir comment faire la configuration des IOT, leurs sécurisations et la communication entre eux.donc la tache a été réussie.